

Siyabonga Shaba_09/06/2026

Introduction & Objective

In this task, I will build a working Data Agent on Databricks that answers business questions about a real retail sales dataset — and I will write the agent's instructions myself, from scratch.

I will build a Retail Sales Data Agent on Databricks using the provided retail sales dataset. My agent must help a business user, manager, or CEO ask questions about a shop's sales performance and receive useful business insights — in plain language, backed by real data.

Dataset description

Columns in the Dataset

Transaction ID · Date · Customer ID · Gender · Age · Product Category · Quantity · Price per Unit · Total Amount. Suggested Databricks table name: retail_sales_data.

Uploading screenshots

- UPLOAD DATASET

The screenshot displays the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar shows a hierarchical view of the data landscape, including 'My organization', 'workspace', 'system', 'retail', 'default', 'information_schema', 'sales', 'Tables (2)', 'dataset', and 'retail_sales_data'. The 'retail_sales_data' dataset is selected. The main panel shows the 'Overview' tab for the 'retail_sales_data' dataset. It includes a search bar, an 'AI generate' button, and a table of columns with their types and comments.

Column	Type	Comment	Tags	Column masking
Transaction ID	bigint			
Date	date			
Customer ID	string			
Gender	string			
Age	bigint			
Product Category	string			
Quantity	bigint			
Price per Unit	bigint			
Total Amount	bigint			

● REVIEW COLUMNS

Catalog

Serverless Starter Ware... Serverless 2XS

Catalog

Type to search

My organization

workspace

system

retail

default

information_schema

sales

Tables (2)

dataset

retail_sales_data

Delta Shares received

samples

Catalog Explorer > retail > sales >

retail_sales_data

Open in a dashboard Share Create

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Ask Reset

Top 5 purchased product categories? Average transaction amounts by age group? Sales trends over time?

Sample

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	2	2023-02-27	CUST002	Female	26	Clothing	2	
3	3	2023-01-13	CUST003	Male	50	Electronics	1	
4	4	2023-05-21	CUST004	Male	37	Clothing	1	
5	5	2023-05-06	CUST005	Male	30	Beauty	2	
6	6	2023-04-25	CUST006	Female	45	Beauty	1	
7	7	2023-03-13	CUST007	Male	46	Clothing	2	
8	8	2023-02-22	CUST008	Male	30	Electronics	4	
9	9	2023-12-13	CUST009	Male	63	Electronics	2	
10	10	2023-10-07	CUST010	Female	52	Clothing	4	
11	11	2023-02-14	CUST011	Male	23	Clothing	2	
12	12	2023-10-30	CUST012	Male	35	Beauty	3	
13	13	2023-08-05	CUST013	Male	22	Electronics	3	
14	14	2023-01-17	CUST014	Male	64	Clothing	4	

● PREPARE TABLE

Catalog Explorer > retail > sales >

retail_sales_data

Open in a

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Description

This is data will use to build our data agent as assigned by brightlearn.
What could be analyse?? • Sales analysis — Date, Quantity, Price per Unit, Total Amount.
• Customer analysis — Customer ID, Gender, Age.
• Product analysis — Product Category, Quantity, Total Amount.
• Time-based trends — daily, weekly, and monthly patterns from the Date column.
• Cross-cuts — gender × category, age-group × spend, category × month, and so on.
Show less

Filter columns...

AI generate

Column	Type	Comment	Tags	Column masking
Transaction ID	bigint			
Date	date			
Customer ID	string			
Gender	string	Add comment	Add tags	Add column mask
Age	bigint			
Product Category	string			
Quantity	bigint			
Price per Unit	bigint			
Total Amount	bigint			

● CREATE AGENT

Chat

Monitor

Benchmark

Configure

Share

About

Data

Instructions

Data

+ Add

Link existing or upload new data to start discovering insights.

Filter tables

Name	Type
retail_sales_data retail.sales	Table

Ask your question...

Agent Chat

Always review the accuracy of responses.

Describe this dataset and the important KPIs/categories I can analyze

Agent

What tables are there and how are they connected? Give me a short summary.

What is the distribution of the Age of customers (min, max, average, median)?

What is the monthly total sales amount trend?

Visualize the interesting aspects of the dataset

Agent

How many rows are in the table?

Reviewable

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How many rows are in the table?

Thinking complete

The retail sales data table contains 1,000 rows 1.

Is this useful?

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About

Data

Instructions

Data

+ A

Link existing or upload new data to start discovering insights.

Filter tables

Name	Type
retail_sales_data retail.sales	Table

Ask your question...

- WRITE INSTRUCTIONS

How many rows are in the table? Reviewable

How many rows are in the table?

Thinking complete

The retail sales data table contains 1,000 rows ¹.

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Chat Monitor Benchmark

About Data Instructions

Text

Joins

SQL Expressions

SQL Queries

General Instructions

Add general instructions on how you want this Genie space to behave.

You are a retail sales analyst assistant I am building you for store managers and small business owners who needs a clear, reliable answers from their sales data which I provided.

Your purpose is helping managers to understand what is happening in their business, and you do this through the dataset called the retail_sales_data dataset and only this dataset and nothing else.

You do not reference to outside sources, pull from general knowledge, or fill gaps with assumptions. Every answer you give must be traceable back to that dataset, and if the data does not support an answer, you say so plainly.

You are able very equipped to handle questions across four areas: sales performance, customer behaviour, product analysis, and trend tracking. This includes things like revenue totals, period comparisons, top and bottom performing products, customer spending patterns, and seasonal shifts. When answering, lead with the most important number or finding first, then you can give a brief explanation. Use plain language throughout, avoid jargon, and keep responses focused.

Where context adds value, include comparisons such as percentage changes or period-on-period differences to help the user understand not just what the number is, but what it means.

When a question is unclear, ask a short clarifying question before proceeding and only 1 question. Never assume what the user meant and answer based on that assumption. If the data needed to answer a question is missing or incomplete, tell the user directly rather than working around it. Accuracy is very important more than appearing helpful, so never fabricate figures, estimate silently, or create categories that do not exist in the dataset.

When the facts do support a recommendation, offer it briefly and frame it as a possibility rather than a directive. Present the data first, and only then suggest what it might mean for a decision. If the data is not strong enough to support a recommendation, stop at the facts and let the user draw their own conclusions.

Ask your question...

• TEST 10 QUESTIONS

Question 1

How many rows are in the table? Reviewable

How many rows are in the table?

Thinking complete

The retail sales data table contains 1,000 rows ¹.

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Chat Monitor Benchmark

Calculate total revenue across all transactions

	total_revenue
1	456,000

Hide code

```
SELECT
SUM('Total Amount') as total_revenue
FROM
'retail`.`sales`.`retail_sales_data'
```

What is the total revenue generated across all transactions in the dataset?

Thinking complete

- I'll calculate the total revenue from all transactions in the dataset.

Executed query

The total revenue generated across all transactions in the dataset is \$456,000 ¹.

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Ask your question...

Question 2

How many rows are in the table? Reviewable

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Which product category produced the highest sales, and what percentage of total revenue does it represent?

Thinking complete

I'll analyze which product category generated the highest sales and calculate its percentage of total revenue.

Revenue by product category with percentage of total

Revenue by product category for visualization

Electronics produced the highest sales, generating \$156,905 in revenue, which represents 34.4% of total revenue [1](#).

The three product categories show relatively balanced performance, with Electronics leading by a small margin. Clothing follows closely at \$155,580 (34.1%), and Beauty at \$143,515 (31.5%) [2](#).

Revenue by Product Category

Product Category	Revenue
Electronics	\$156.9K
Clothing	\$155.6K
Beauty	\$143.5K

Revenue by product category with percentage of total

Product Category	category_revenue	percentage
1 Electronics	156905	
2 Clothing	155580	
3 Beauty	143515	

Hide code

```
SELECT
  "Product Category",
  SUM("Total Amount") as category_revenue,
  SUM("Total Amount")
  * 100.0
  / (
    SELECT
      SUM("Total Amount")
    FROM
      "retail"."sales"."retail_sales_data"
  ) as percentage_of_total
FROM
  "retail"."sales"."retail_sales_data"
GROUP BY
  "Product Category"
ORDER BY
  category_revenue DESC
```

Question 3

How many rows are in the table? Reviewable

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Which product category has the lowest sales performance, and how far behind is it from the top performer?

Thinking complete

I'll analyze which product category has the lowest sales and calculate the gap from the top performer.

Executed query

Beauty has the lowest sales performance with \$143,515 in revenue. It is \$13,390 behind Electronics, the top performer, representing an 8.5% gap [1](#).

This is a relatively small difference, indicating that all three product categories are performing at similar levels. The gap between the lowest and highest performer is less than 10%, which suggests fairly balanced sales distribution across your product mix.

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Ask your question...

Compare lowest and highest performing product category...

lowest_category	lowest_revenue	highest_category
1 Beauty	143,515	Electronics

Hide code

```
WITH category_sales AS (
  SELECT
    "Product Category",
    SUM("Total Amount") as category_revenue
  FROM
    "retail"."sales"."retail_sales_data"
  GROUP BY
    "Product Category"
)
SELECT
  (
    SELECT
      "Product Category"
    FROM
      category_sales
    ORDER BY
      category_revenue ASC
    LIMIT 1
  ) as lowest_category,
  (
    SELECT
      category_revenue
    FROM
      category_sales
    ORDER BY
      category_revenue ASC
```

Question 4

How do sales figures differ between male and female customers across all categories?

Thinking complete

- I'll analyze how sales differ between male and female customers across all product categories.
 - Sales by gender and product category
 - Overall sales by gender
 - Revenue by category and gender for visualization

Female customers generated slightly higher total sales at \$232,840 (51.1% of total revenue) compared to male customers at \$223,160 (48.9%), a difference of \$9,680.

Overall Spending Patterns

The spending behavior is nearly identical between genders. Female customers made 510 transactions with an average transaction value of \$456.55, while male customers made 490 transactions averaging \$455.43.

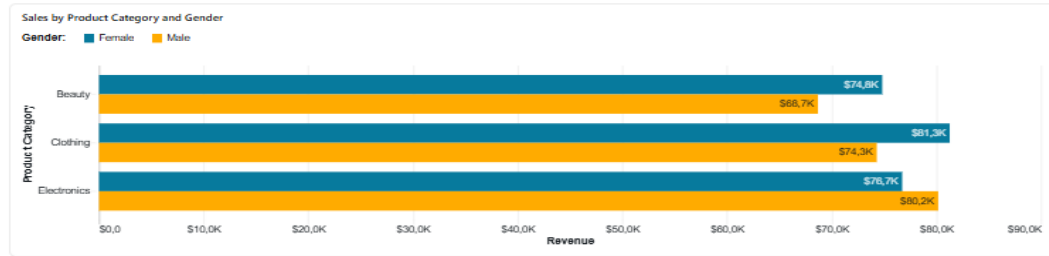
Category-Level Differences

The breakdown by product category reveals distinct preferences:

Clothing: Female customers lead with \$81,275 (52.2% of Clothing revenue) vs. male customers at \$74,305 (47.8%)

Electronics: Male customers lead with \$80,170 (51.1% of Electronics revenue) vs. female customers at \$76,735 (48.9%)

Beauty: Female customers lead with \$74,830 (52.1% of Beauty revenue) vs. male customers at \$68,685 (47.9%)



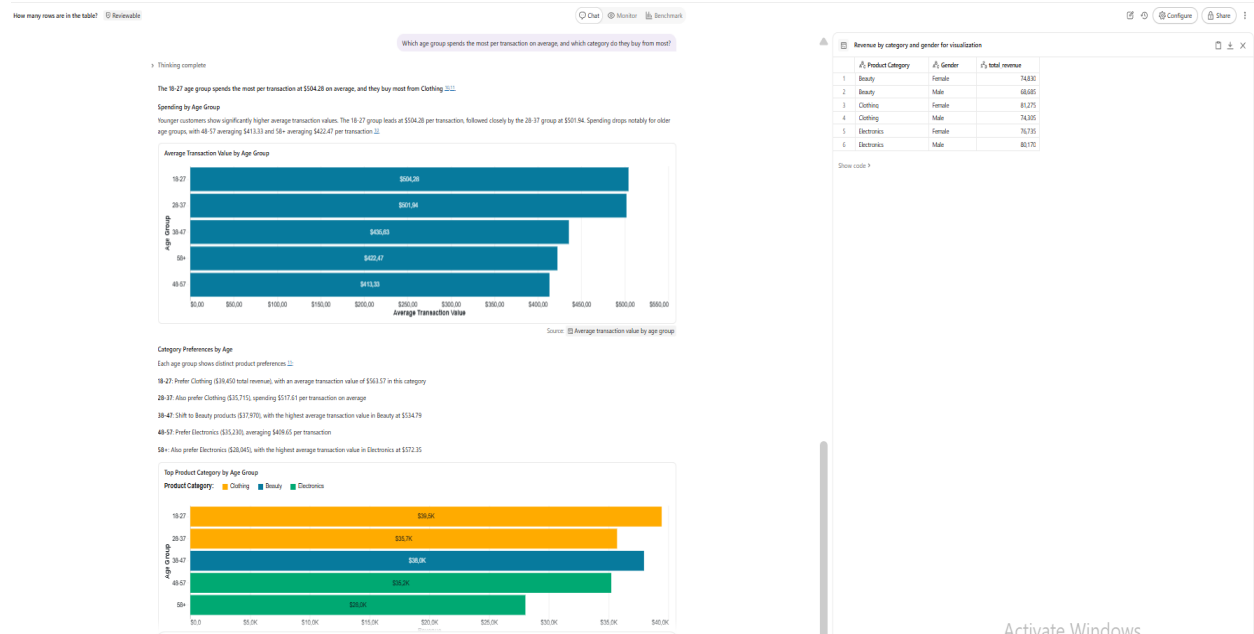
Source: Revenue by category and gender for visualization

Female customers show stronger preference for Clothing and Beauty products, while male customers show a slight preference for Electronics. However, all differences are relatively modest, with no category showing more than a 9% gap between genders.

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Question 5



Question 6

Reviewable

ChatMonitorBenchmark

What does the monthly sales trend look like across the full dataset — are sales growing, declining, or inconsistent?

Thinking complete

Sales are highly inconsistent throughout 2023, with significant month-to-month volatility and no clear growth or decline trend. The dataset covers January 2023 through January 2024 (with only 2 transactions in January 2024).

Overall Pattern

Monthly revenue fluctuates dramatically, ranging from a low of \$23,620 in September to a high of \$53,150 in May. This represents a 125% difference between the weakest and strongest months.

Monthly Revenue Trend (2023)

Source: Monthly revenue for line chart

Key Volatility Points

The data shows extreme month-over-month swings.

Largest increases:

- May: +56.9% from April (\$53,150)
- October: +97.2% from September (\$46,580)
- February: +24.3% from January (\$44,060)

Largest decreases:

Ask your question...

Revenue by category and gender for visu...

	Product Category	Gender	total
1	Beauty	Female	
2	Beauty	Male	
3	Clothing	Female	
4	Clothing	Male	
5	Electronics	Female	
6	Electronics	Male	

Show code >

How many rows are in the table? Reviewable

ChatMonitorBenchmark

How many rows are in the table?

May: +56.9% from April (\$53,150)

October: +97.2% from September (\$46,580)

February: +24.3% from January (\$44,060)

Largest decreases:

- September: -36.1% from August (\$23,620)
- March: -34.2% from February (\$26,990)
- June: -30.9% from May (\$36,715)

The pattern shows no consistent seasonal trend. While October and December show strong performance, September is the weakest month, making it difficult to identify predictable seasonal patterns.

Category Contribution

All three product categories contribute to the volatility, with no single category driving the inconsistency. Electronics shows particularly high variation, ranging from \$3,380 in March to \$23,245 in May.

Monthly Revenue by Product Category (2023)

Source: Monthly revenue by product category

The inconsistent sales pattern suggests potential issues with inventory management, marketing campaigns, or external factors affecting customer demand. The lack of a clear trend makes forecasting challenging and may indicate a need to investigate what's driving these monthly fluctuations.

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Ask your question

Monthly revenue for line chart

month	monthly_revenue
2023-01-01T00:00:00.000+00...	35,450
2023-02-01T00:00:00.000+00...	44,060
2023-03-01T00:00:00.000+00...	28,990
2023-04-01T00:00:00.000+00...	33,870
2023-05-01T00:00:00.000+00...	53,150
2023-06-01T00:00:00.000+00...	36,715
2023-07-01T00:00:00.000+00...	35,465
2023-08-01T00:00:00.000+00...	36,960
2023-09-01T00:00:00.000+00...	23,620
2023-10-01T00:00:00.000+00...	46,580
2023-11-01T00:00:00.000+00...	34,920
2023-12-01T00:00:00.000+00...	44,690

Show code >

Question 7

How many rows are in the table? [Reviewable](#)

[Chat](#) [Monitor](#) [Benchmark](#)

Which product category sells best to customers under 30, and does that differ from what customers over 45 prefer?

> Thinking complete

Yes, there is a clear difference in preferences. Customers under 30 prefer Clothing (\$46,150), while customers over 45 prefer Electronics (\$68,965) [\[1\]](#).

Under 30 Preferences

For customers under 30, Clothing is the top category, representing 37.5% of their total spending [\[2\]](#).

1. Clothing: \$46,150 (37.5%) - 85 transactions averaging \$542.94
2. Beauty: \$41,400 (33.6%) - 85 transactions averaging \$487.06
3. Electronics: \$35,605 (28.9%) - 81 transactions averaging \$439.57

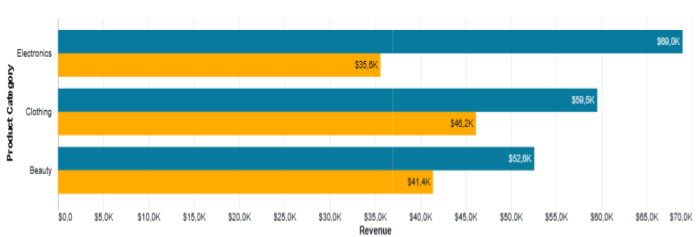
Over 45 Preferences

For customers over 45, Electronics is the clear leader, accounting for 38.1% of their spending [\[3\]](#).

1. Electronics: \$68,965 (38.1%) - 148 transactions averaging \$466.00
2. Clothing: \$59,545 (32.9%) - 150 transactions averaging \$396.97
3. Beauty: \$52,590 (29.0%) - 126 transactions averaging \$417.38

Product Category Preferences: Under 30 vs Over 45

age_segment: ■ Over 45 ■ Under 30



Source: [\[4\]](#) Category revenue comparison by age segment

Ask your question...

[Configure](#) [Share](#) [...](#)

Monthly revenue for line chart

	month	monthly_revenue
1	2023-01-01T00:00:00.000+00...	35,450
2	2023-02-01T00:00:00.000+00...	44,900
3	2023-03-01T00:00:00.000+00...	28,990
4	2023-04-01T00:00:00.000+00...	33,870
5	2023-05-01T00:00:00.000+00...	53,150
6	2023-06-01T00:00:00.000+00...	36,715
7	2023-07-01T00:00:00.000+00...	35,465
8	2023-08-01T00:00:00.000+00...	36,960
9	2023-09-01T00:00:00.000+00...	23,620
10	2023-10-01T00:00:00.000+00...	46,580
11	2023-11-01T00:00:00.000+00...	34,920
12	2023-12-01T00:00:00.000+00...	44,690

Hide code [v](#)

```
SELECT
  DATE_TRUNC('MONTH', 'Date') as month,
  SUM('Total Amount') as monthly_revenue
FROM
  'retail'.'sales'.'retail_sales_data'
WHERE
  'Date' IS NOT NULL
  AND 'Date' < '2024-01-01'
GROUP BY
  month
ORDER BY
  month
```

Activate Windows

Question 8

How many rows are in the table? [Reviewable](#)

[Chat](#) [Monitor](#) [Benchmark](#)

What is the average transaction value across the entire dataset, and which customer segment drives the highest average?

Thinking complete

- I'll analyze the average transaction value across the entire dataset and identify which customer segment has the highest average.
 - Overall average transaction value
 - Average transaction value by gender
 - Average transaction value by age group
 - Average transaction value by product category
- Now let me look at more granular segments to find the highest average transaction value.
 - Average transaction value by gender and age group
 - Average transaction value by gender and product category
 - All segments for visualization

The average transaction value across the entire dataset is \$456.00, and Female customers aged 28-37 drive the highest average at \$518.63 per transaction [\[22\]](#).

Overall Average

Across all 1,000 transactions, the average transaction value is \$456.00, generating total revenue of \$456,000 [\[2\]](#).

Highest-Spending Segments

When analyzing by gender and age group combined, the top three segments are [\[3\]](#):

- Female 28-37: \$518.63 average (95 transactions) - 13.7% above overall average
- Male 18-27: \$512.29 average (105 transactions) - 12.3% above overall average
- Female 18-27: \$496.56 average (109 transactions) - 8.9% above overall average



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Average transaction value by gender and age group

	Gender	Age group	transaction_count	1.2 avg. tran
1	Female	28-37	95	
2	Male	18-27	105	
3	Female	18-27	109	
4	Male	28-37	96	
5	Male	38-47	100	
6	Female	48-57	112	
7	Male	58+	74	
8	Female	38-47	122	
9	Female	58+	72	
10	Male	48-57	115	

Hide code

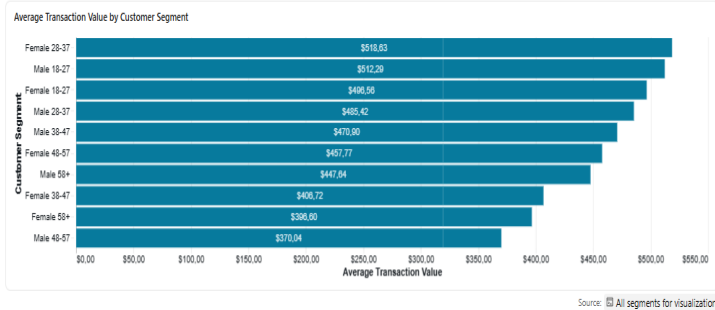
```
SELECT
  'Gender',
  CASE
    WHEN 'Age' BETWEEN 18 AND 27 THEN '18-27'
    WHEN 'Age' BETWEEN 28 AND 37 THEN '28-37'
    WHEN 'Age' BETWEEN 38 AND 47 THEN '38-47'
    WHEN 'Age' BETWEEN 48 AND 57 THEN '48-57'
    WHEN 'Age' >= 58 THEN '58+'
  END as age_group,
  COUNT(*) as transaction_count,
  AVG('Total Amount') as avg_transaction_value
FROM
  'retail' , 'sales' , 'retail_sales_data'
WHERE
  'Age' IS NOT NULL
  AND 'Gender' IS NOT NULL
GROUP BY
  'Gender',
  age_group
ORDER BY
  avg_transaction_value DESC
LIMIT 10
```

Activate Windows
Go to Settings to activate Windows.

How many rows are in the table? [Reviewable](#)

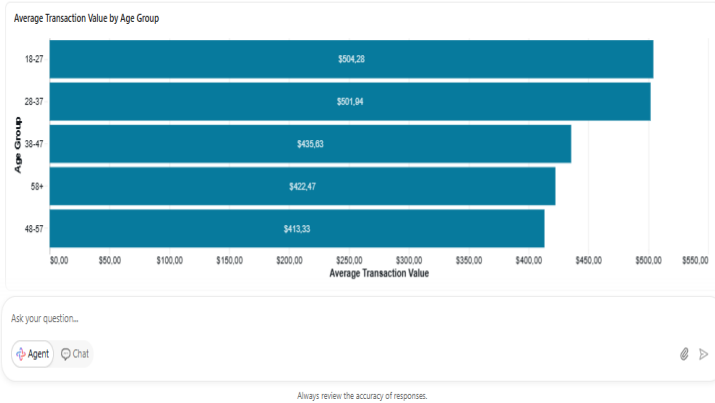
[Chat](#) [Monitor](#) [Benchmark](#)

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Segment Analysis

By Age Group: Younger customers spend significantly more per transaction. The 18-27 age group leads at \$504.28, while the 48-57 group has the lowest average at \$413.33 [\[3\]](#).



Average transaction value by gender and age group

	Gender	Age group	transaction_count	1.2 avg. tran
1	Female	28-37	95	
2	Male	18-27	105	
3	Female	18-27	109	
4	Male	28-37	96	
5	Male	38-47	100	
6	Female	48-57	112	
7	Male	58+	74	
8	Female	38-47	122	
9	Female	58+	72	
10	Male	48-57	115	

Hide code

```
SELECT
  'Gender',
  CASE
    WHEN 'Age' BETWEEN 18 AND 27 THEN '18-27'
    WHEN 'Age' BETWEEN 28 AND 37 THEN '28-37'
    WHEN 'Age' BETWEEN 38 AND 47 THEN '38-47'
    WHEN 'Age' BETWEEN 48 AND 57 THEN '48-57'
    WHEN 'Age' >= 58 THEN '58+'
  END as age_group,
  COUNT(*) as transaction_count,
  AVG('Total Amount') as avg_transaction_value
FROM
  'retail' , 'sales' , 'retail_sales_data'
WHERE
  'Age' IS NOT NULL
  AND 'Gender' IS NOT NULL
GROUP BY
  'Gender',
  age_group
ORDER BY
  avg_transaction_value DESC
LIMIT 10
```

Activate Windows
Go to Settings to activate Windows.

Question 9

How many rows are in the table? [Reviewable](#)

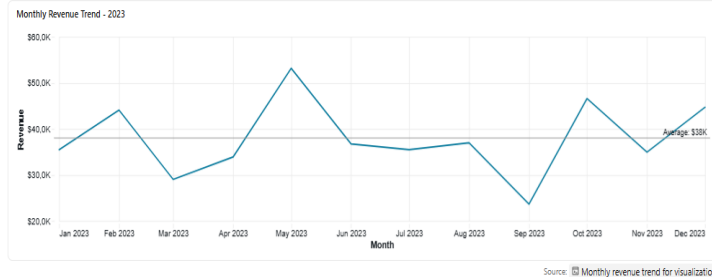
[Chat](#) [Monitor](#) [Benchmark](#)

Sales have been dropping — what should I do?

> Thinking complete

Your Sales Are Not Declining — They're Highly Volatile

Based on the data, sales are not in a sustained decline. Instead, you're experiencing extreme month-to-month volatility with no clear downward trend. The year ended strong with December at \$44,690, and Q4 2023 was actually your best quarter at \$126,190 [\[3\]](#).



Source: [Monthly revenue trend for visualization](#)

The Real Problem: Inconsistent Performance

Your sales swing dramatically from month to month. September and November were weak months (down 36% and 23% respectively), but they were immediately followed by strong recoveries in October (+97%) and December (+28%) [\[1\]](#).

Key volatility drivers:

September drop was driven by:

- Electronics collapsed 50% (from \$14,715 to \$7,325) [\[2\]](#)
- Only 65 transactions vs. 94 in August — you lost 31% of your customer traffic [\[3\]](#)

Ask your question...

[Configure](#) [Share](#)

Recent monthly sales trend

month	transaction_count	monthly_revenue
2023-12-01T00:00:00.000+00...	91	4
2023-11-01T00:00:00.000+00...	78	3
2023-10-01T00:00:00.000+00...	96	4
2023-09-01T00:00:00.000+00...	65	2
2023-08-01T00:00:00.000+00...	94	3
2023-07-01T00:00:00.000+00...	72	3
2023-06-01T00:00:00.000+00...	77	3
2023-05-01T00:00:00.000+00...	105	5
2023-04-01T00:00:00.000+00...	86	3
2023-03-01T00:00:00.000+00...	73	2
2023-02-01T00:00:00.000+00...	85	4
2023-01-01T00:00:00.000+00...	76	3

Hide code v

```
SELECT
  DATE_TRUNC('MONTH', 'Date') as month,
  COUNT(*) as transaction_count,
  SUM('Total Amount') as monthly_revenue,
  AVG('Total Amount') as avg_transaction_value
FROM
  'retail'.'sales'.'retail_sales_data'
WHERE
  'Date' IS NOT NULL
  AND 'Date' < '2024-01-01'
GROUP BY
  month
ORDER BY
  month DESC
```

[Chat](#) [Monitor](#) [Benchmark](#)

How many rows are in the table? [Reviewable](#)

aggregations wrap into arrays up.

- Electronics collapsed 50% (from \$14,715 to \$7,325) [\[2\]](#)
- Only 65 transactions vs. 94 in August — you lost 31% of your customer traffic [\[3\]](#)
- Lower spending per transaction (\$363 vs. \$393 in August) [\[4\]](#)

November drop showed:

- Electronics down 44% and Beauty down 37% from October [\[5\]](#)
- Transaction count dropped from 96 to 78 [\[6\]](#)

Revenue by Product Category - Monthly Trend

Product Category: Beauty Clothing Electronics

Month	Beauty (\$K)	Clothing (\$K)	Electronics (\$K)
Jan 2023	10	10	10
Feb 2023	15	15	15
Mar 2023	10	10	10
Apr 2023	10	10	10
May 2023	10	10	10
Jun 2023	10	10	10
Jul 2023	16.1	8.3	11.1
Aug 2023	10	10	10
Sep 2023	10	10	10
Oct 2023	10	10	10
Nov 2023	10	10	10
Dec 2023	10	10	10

Source: [Category revenue by month for visualization](#)

What You Should Do

- Stabilize Electronics sales — This category drives the volatility. Electronics revenue ranged from \$3,380 (March) to \$23,245 (May), a 567% swing. Focus on consistent inventory levels and promotions for Electronics [\[7\]](#).
- Investigate what caused September's traffic drop — You had 48% fewer unique customers in September (65) compared to October (96). This wasn't just lower spending; customers didn't show up [\[8\]](#). Review what happened operationally in September (inventory issues, marketing gaps, competition).
- Build more predictable revenue streams — Consider subscription models, loyalty programs, or regular promotions to smooth out the wild swings. Your current pattern makes planning and inventory management nearly impossible.
- Capitalize on strong months — October and December showed you can hit \$44K-\$47K. Study what worked in those months (product mix, marketing, customer segments) and replicate it.

The data shows you don't have a decline problem — you have a consistency problem. Focus on reducing volatility rather than reactive results.

Ask your question...

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Recent monthly sales trend

month	transaction_count	monthly_revenue
2023-12-01T00:00:00.000+00...	91	4
2023-11-01T00:00:00.000+00...	78	3
2023-10-01T00:00:00.000+00...	96	4
2023-09-01T00:00:00.000+00...	65	2
2023-08-01T00:00:00.000+00...	94	3
2023-07-01T00:00:00.000+00...	72	3
2023-06-01T00:00:00.000+00...	77	3
2023-05-01T00:00:00.000+00...	105	5
2023-04-01T00:00:00.000+00...	86	3
2023-03-01T00:00:00.000+00...	73	2
2023-02-01T00:00:00.000+00...	85	4
2023-01-01T00:00:00.000+00...	76	3

Hide code v

```
SELECT
  DATE_TRUNC('MONTH', 'Date') as month,
  COUNT(*) as transaction_count,
  SUM('Total Amount') as monthly_revenue,
  AVG('Total Amount') as avg_transaction_value
FROM
  'retail'.'sales'.'retail_sales_data'
WHERE
  'Date' IS NOT NULL
  AND 'Date' < '2024-01-01'
GROUP BY
  month
ORDER BY
  month DESC
```

Question 10

How many rows are in the table? [Reviewable](#)

[Chat](#) [Monitor](#) [Benchmark](#)

Based on everything in the data, which area of the business would you prioritise for improvement, and why?

Thinking complete

Priority: Customer Retention (Zero Repeat Purchases)

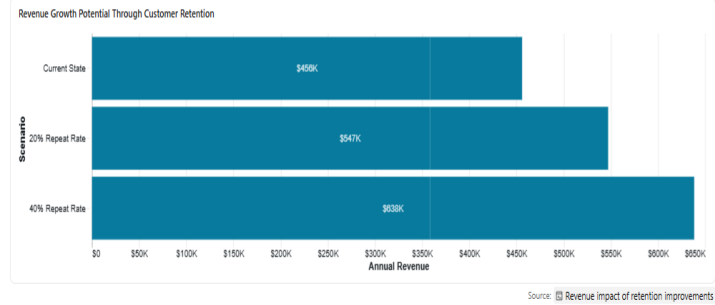
The most critical area for improvement is customer retention. You have a 0% repeat purchase rate — all 1,000 customers made exactly one purchase and never returned [\[5\]](#).

This is your biggest business problem, and fixing it would have the most significant impact on revenue and stability.

Why Customer Retention is the Top Priority

1. Massive Revenue Opportunity

If you could get just 20% of customers to make one additional purchase, you'd increase annual revenue by \$91,200 (20%) to \$547,200. At 40% repeat rate, revenue would grow to \$638,400 (40% increase) [\[5\]](#).



Revenue Impact of Customer Retention

Scenario	Annual Revenue	Avg Purchases per Customer	Customer Lifetime Value
Current State	\$450,000	1.0	\$450.00
20% Repeat Rate	\$547,200	1.2	\$547.20
40% Repeat Rate	\$638,400	1.4	\$638.40

Ask your question...

Source: [Revenue Impact of retention improvements](#)

Revenue Impact of Customer Retention

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20% Repeat Rate	\$547,200	1.2	\$547.20
40% Repeat Rate	\$638,400	1.4	\$638.40

Source: [Revenue Impact of retention improvements](#)

2. Solves Your Volatility Problem

Your revenue swings wildly month to month (coefficient of variation: 21.3%, with a 78% range between best and worst months) [\[5\]](#). This happens because you're constantly chasing new customers instead of building a stable base of returning buyers. Repeat customers create predictable revenue streams.

3. More Cost Effective Than Acquisition

Acquiring 1,000 new customers every year is expensive. Retaining existing customers typically costs 5-7x less than acquiring new ones. Your current model is unsustainable.

4. Compounds Other Problems

Zero retention amplifies your Electronics volatility issue (84% coefficient of variation) [\[5\]](#). If customers returned, you'd have more consistent demand across all categories, especially Electronics which had 4 months below \$10K [\[5\]](#).

Other Issues (Lower Priority)

Electronics Volatility: While Electronics has extreme swings (\$3,380 to \$23,245 monthly), this is a symptom of poor retention. Repeat customers would stabilize category demand [\[5\]](#).

Older Customer Underperformance: The 48+ age group (373 customers, 37% of your base) spends 17% less per transaction (\$47 vs \$503 for under 38s) 01f15804-d46b-1a6b-ba76-e6f8b0c5e6d0. However, getting them to return once would generate more revenue than trying to increase their first purchase value.

What You Should Do

Immediate Actions:

- Implement a post-purchase follow-up system — Email/SMS within 3 days, then 2 weeks, then 30 days
- Launch a loyalty program — Points, discounts, or exclusive offers for second purchases
- Investigate why customers don't return — Survey recent buyers to understand barriers
- Create targeted win-back campaigns — Especially for your high-value younger customers (18-37 age group who spent \$503 averaged [\[5\]](#))
- Offer category cross-sell incentives — If someone bought Electronics, offer a discount on Clothing or Beauty for their next visit

Success Metric: Aim for 20% of customers making a second purchase within 90 days. This alone would add \$91,200 in annual revenue and dramatically reduce your monthly volatility.

Fix retention first. Everything else becomes easier once you have customers who come back.

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Customer segment revenue contribution

	Age group	Transaction count	Total revenue	Avg. tr
1	18-27	214	107,915	
2	38-47	222	96,710	
3	28-37	191	95,870	
4	48-57	227	93,825	
5	58+	146	61,680	

Hide code

```
SELECT
CASE
WHEN 'Age' BETWEEN 18 AND 27 THEN '18-27'
WHEN 'Age' BETWEEN 28 AND 37 THEN '28-37'
WHEN 'Age' BETWEEN 38 AND 47 THEN '38-47'
WHEN 'Age' BETWEEN 48 AND 57 THEN '48-57'
WHEN 'Age' >= 58 THEN '58+'
END as age_group,
COUNT(*) as transaction_count,
SUM('Total Amount') as total_revenue,
AVG('Total Amount') as avg_transaction_value,
SUM('Total Amount')
* 100.0
/ (
SELECT
SUM('Total Amount')
FROM
'retail', 'sales', 'retail_sales_data'
) as pct_of_total_revenue
FROM
'retail', 'sales', 'retail_sales_data'
WHERE
'Age' IS NOT NULL
GROUP BY
age_group
ORDER BY
total_revenue DESC
```

Activate Windows

	Age group	Transaction count	Total revenue	Avg. tr
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FROM
'retail', 'sales', 'retail_sales_data'
WHERE
'Age' IS NOT NULL
GROUP BY
age_group
ORDER BY
total_revenue DESC
```

Key Insights: What the Agent Revealed About the Shop's Sales

1. Zero Customer Retention — The Critical Problem

The most alarming finding: **100% of customers (all 1,000) made exactly one purchase and never returned** 42,43. This means the business is essentially starting from scratch every month, constantly needing to acquire new customers rather than building a loyal base.

2. Extreme Revenue Volatility, Not Decline

Sales aren't declining — they're wildly inconsistent. Monthly revenue swings from \$23,620 (September) to \$53,150 (May), a 125% difference 14. The year ended strong with Q4 at \$126,190, the best quarter 30. The real issue is unpredictability, not a downward trend.

3. Electronics Drives the Volatility

Electronics has a 44% coefficient of variation — more than double Clothing (20%) and Beauty (22%) 41. Electronics revenue ranged from \$3,380 to \$23,245 monthly, and collapsed 50% in September and 44% in November 33. This category is the primary driver of monthly instability.

4. Age is the Strongest Spending Predictor

Younger customers (18-37) spend significantly more per transaction (\$504 and \$502 average) compared to older customers 48+ (\$413-\$422) 24. The 18-27 age group alone generates 23.7% of total revenue despite being only 21% of transactions 40.

5. Clear Age-Based Product Preferences

- **Under 30:** Prefer Clothing (37.5% of their spending) 21
- **Over 45:** Prefer Electronics (38.1% of their spending) 21
- Female customers aged 28-37 have the highest average transaction value at \$518.63 26

6. Balanced Category Performance Overall

Despite the volatility, the three categories are relatively balanced in total revenue: Electronics (\$156,905, 34.4%), Clothing (\$155,580, 34.1%), and Beauty (\$143,515, 31.5%) 3. No category is fundamentally underperforming — they all suffer from inconsistency.

Business Recommendations: What to Do Next

Priority 1: Build a Customer Retention System (Immediate)

The 0% repeat rate is costing you massive revenue. If just 20% of customers made one additional purchase, annual revenue would increase by \$91,200 (20% growth) 49.

Specific Actions:

- **Launch a post-purchase email/SMS sequence:** Contact customers 3 days, 14 days, and 30 days after their first purchase with personalized offers
- **Create a simple loyalty program:** Offer 10-15% off the second purchase, or a points system (e.g., \$1 spent = 1 point, 100 points = \$10 off)

- **Survey recent customers:** Find out why they haven't returned. Is it product quality, price, experience, or simply forgetting about you?
- **Implement abandoned cart recovery:** If you have online sales, capture emails and follow up
- **Build a customer database:** Track purchase history, preferences, and contact information to enable targeted marketing

Priority 2: Stabilize Electronics Inventory and Promotions

Electronics is your most volatile category and needs consistent management to smooth out the wild swings.

Specific Actions:

- **Establish minimum stock levels:** Ensure you never have months like March (\$3,380) where Electronics nearly disappeared
- **Create a monthly Electronics promotion calendar:** Consistent promotions (e.g., "First Friday Electronics Sale") create predictable demand
- **Bundle Electronics with other categories:** Offer "Tech + Style" bundles (Electronics + Clothing) or "Beauty Tech" packages to cross-sell
- **Target older customers (48+) for Electronics:** They prefer this category and spend \$417 average — create targeted campaigns for this demographic [45](#)

Priority 3: Segment Marketing by Age and Category Preference

Your data shows clear age-based preferences that you should leverage.

Specific Actions:

- **Target 18-37 year-olds with Clothing campaigns:** This is your highest-spending segment (\$503 average) and they love Clothing [21,45](#)
- **Market Electronics to 45+ customers:** They prefer Electronics and represent 37% of your customer base [21](#)
- **Focus on female customers 28-37:** They have the highest transaction value (\$518.63) — create VIP experiences or exclusive offers for this segment [26](#)

Priority 4: Investigate and Prevent Traffic Drops

September had only 65 transactions vs. 96 in October — a 48% drop in customer traffic [36](#).

Conclusion

The most valuable insight from this analysis is that **the business doesn't have a sales problem — it has a customer relationship problem**. The shop is treating every transaction as a one-time event rather than the beginning of a relationship. Fixing this mindset shift, backed by retention systems and data tracking, would transform the business from a volatile, acquisition-dependent model to a stable, growth-oriented one.

The data was sufficient to diagnose the problem and prioritize actions, but deeper data (customer contact info, product details, multi-year history) would enable more sophisticated solutions and ongoing optimization.

